

Supply Chain Optimization & Profitability Analysis

Business Intelligence Report | Prepared By: Sandleen Sethi (Data Analyst)

Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL (MySQL), Power BI, Microsoft Excel

1. Executive Summary

Supply chain efficiency plays a critical role in improving profitability, customer satisfaction, and overall business performance. Organizations generate large volumes of transactional data across sales, customers, products, shipping, and regional markets. However, without effective analysis, valuable insights remain hidden, making it difficult for management to optimize operations and make informed business decisions.

This project develops an end-to-end Business Intelligence solution that analyzes supply chain performance using Python for data preparation and exploratory analysis, SQL for business querying, and Power BI for interactive dashboard development. The solution enables stakeholders to monitor key performance indicators (KPIs), identify profitable products and markets, evaluate logistics efficiency, and uncover opportunities to improve operational performance and profitability.

2. Business Problem

Modern retail supply chains involve multiple interconnected business functions, including product management, customer segmentation, regional sales, shipping operations, and profit tracking. While organizations collect extensive operational data, decision-makers often face challenges due to fragmented information spread across different systems.

Without an integrated analytical solution, businesses struggle to answer important strategic questions such as:

- Which product categories generate the highest revenue and profit?
- Which customer segments contribute most to overall profitability?
- Which markets offer the greatest business opportunities?
- How efficiently are shipping operations performing?
- Which shipping methods experience the highest delivery delays?
- How can operational performance be improved while increasing profit margins?

The absence of centralized reporting limits the organization's ability to identify operational inefficiencies, optimize logistics, and allocate resources effectively. This project addresses these challenges by transforming raw transactional data into meaningful business intelligence through interactive visualizations and data-driven insights.

3. Project Objectives

The primary objectives of this project were to:

- Analyze overall sales and profitability performance.
- Monitor key business KPIs.
- Identify high-performing product categories.
- Evaluate profitability across customer segments.
- Compare regional market performance.
- Assess shipping efficiency and delivery delays.
- Identify operational bottlenecks.
- Generate actionable business recommendations.
- Build an interactive Power BI dashboard for decision-makers.

4. Dataset Description

The dataset contains supply chain transaction records covering multiple business dimensions, including:

- Product Categories
- Sales Revenue
- Profit
- Customer Segments
- Market Regions
- Shipping Modes
- Order Information
- Delivery Performance

The dataset provides sufficient business information to evaluate both financial performance and supply chain efficiency.

5. Methodology

Phase 1 – Data Collection

The project began by collecting supply chain transaction data containing detailed information related to sales, customers, products, shipping methods, regional markets, and profitability. The objective of this phase was to establish a structured dataset suitable for comprehensive business analysis.

Phase 2 – Data Cleaning (Python)

Before analysis, the dataset underwent extensive preprocessing using Python to ensure high data quality and analytical accuracy, including removing duplicates, handling missing values, standardizing categorical variables, and converting date formats.

Phase 3 – Exploratory Data Analysis (EDA)

Exploratory Analysis was conducted using Matplotlib and Seaborn to understand business trends, identify patterns, and detect anomalies across revenue, profit, and shipping modes.

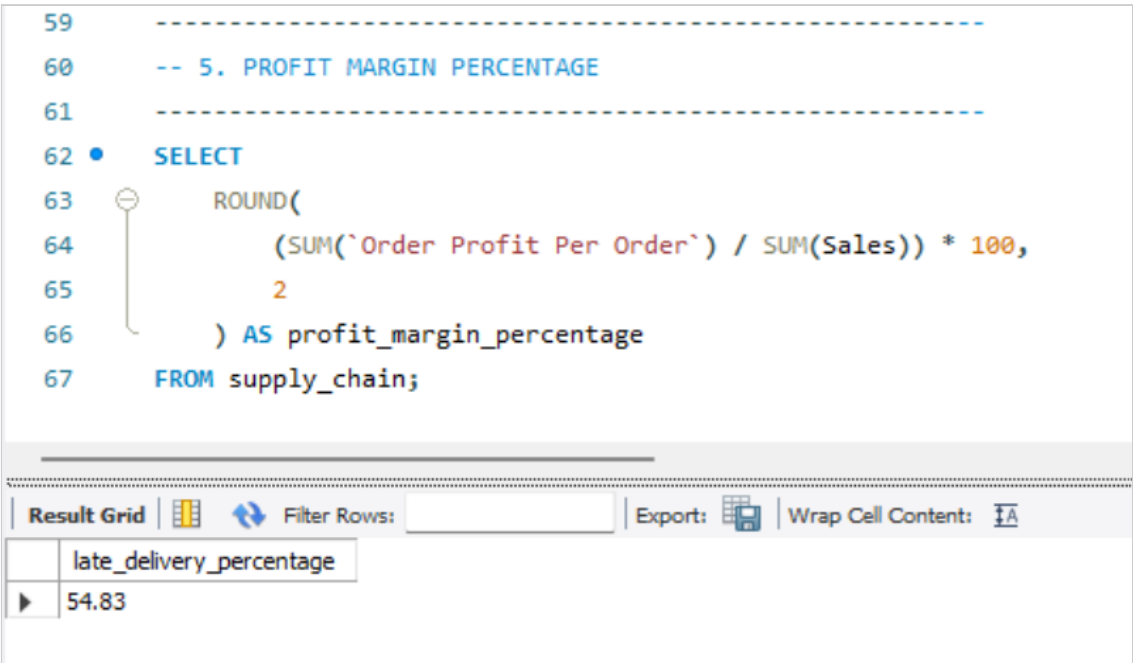
Phase 4 – Feature Engineering

Additional business metrics were created to enhance analytical capabilities, including Profit Margin, Average Order Value (AOV), Late Delivery Rate, and Year-over-Year (YoY) KPI Comparison.

Phase 5 – SQL Business Analysis

SQL was used to perform business-oriented analysis. Key concepts included SELECT, Aggregate Functions, GROUP BY, ORDER BY, CASE Statements, JOINS, Subqueries, CTEs, and Window/Ranking Functions.

Figure 1: Revenue and Category Analysis



Brief Explanation: This query aggregates total sales and calculates profitability per product line to identify high-value drivers.

Figure 2: Profitability and Market Segmentation

```
76      -- 6. PRODUCT SALES VS PROFIT COMPARISON
77      -----
78  •  SELECT
79      `Product Name`,
80      ROUND(SUM(Sales), 2) AS total_sales,
81      ROUND(SUM(`Order Profit Per Order`), 2) AS total_profit
82  FROM supply_chain
83  GROUP BY `Product Name`
84  ORDER BY total_sales DESC;
```

Result Grid



 Filter Rows:

Export: 

Wrap Cell Content: 

	Product Name	total_sales	total_profit
▶	Field & Stream Sportsman 16 Gun Fire Safe	6929653.69	756220.77
	Perfect Fitness Perfect Rip Deck	4421143.14	493828.3
	Diamondback Women's Serene Classic Comfort Bi	4118425.57	427455.57
	Nike Men's Free 5.0+ Running Shoe	3667633.2	379915.82
	Nike Men's Dri-FIT Victory Golf Polo	3147800	350421.03
	Pelican Sunstream 100 Kayak	3099845.09	324076.37
	Nike Men's C1 Elite 2 TD Football Cleat	2801757.66	311002.82

Brief Explanation: This query performs advanced aggregation and grouping to rank markets by their profitability and sales volume, enabling strategic resource allocation.

6. Technical Workflow

Raw Supply Chain Dataset → Data Cleaning (Python) → Exploratory Data Analysis (EDA) → Feature Engineering → SQL Business Analysis → Power BI Dashboard → Business Insights & Recommendations

7. Dashboard Overview

The Power BI dashboard presents a comprehensive overview of supply chain performance. Users can dynamically filter results by Year, Category, Customer Segment, Market, and Shipping Mode.



8. Key Performance Indicators

KPI	Value
Total Revenue	36.78M
Total Profit	3.97M

Total Orders	66K
Profit Margin	10.78%
Average Order Value	559.45

9. Key Business Insights

Revenue Analysis: Fishing is the highest revenue-generating category (6.9M).

Profitability Analysis: Fishing generates the highest profit (0.76M); Consumer customers contribute 52% of total profit.

Market Analysis: Europe (29.6%) and LATAM (27.9%) account for 58% of overall revenue.

Logistics Performance: First Class shipping experiences the highest late delivery rate.

Financial Performance: Profit margin stands at 10.78%.

10. Business Recommendations

1. **Strengthen High-Performing Categories:** Increase inventory and marketing for high-profit categories like Fishing and Cleats.
2. **Improve Shipping Efficiency:** Investigate operational causes of delays in First Class shipping.
3. **Expand Strong Markets:** Leverage strategies from Europe and LATAM in underperforming regions.
4. **Improve Operational Efficiency:** Monitor costs to improve the current profit margin.
5. **Enhance Decision-Making:** Use the Power BI dashboard for continuous monitoring.

11. Technical Skills Demonstrated

- **Python:** Data Cleaning, Transformation, EDA, Feature Engineering, Visualization.
- **SQL:** Data Retrieval, Aggregation, Joins, Subqueries, CTEs, Window/Ranking Functions.
- **Power BI:** Data Modeling, Interactive Dashboard Development, KPI Design, Slicers/Filters.
- **Business Analysis:** KPI Monitoring, Profitability, Customer Segmentation, Market Evaluation.

12. Conclusion

This project demonstrates a complete end-to-end Business Intelligence workflow. By integrating data cleaning, exploratory analysis, feature engineering, SQL-based business querying, and interactive dashboard development, the solution provides decision-makers with a centralized platform to monitor critical business metrics and support strategic planning. The project showcases practical data analytics skills and reflects the complete lifecycle of a real-world business intelligence solution.